

Benchmarking Vision Models in Federated Learning *under data heterogeneity for Medical Image Classification*

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Thesis for Bachelor's Degree in Computer Science

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Introduction

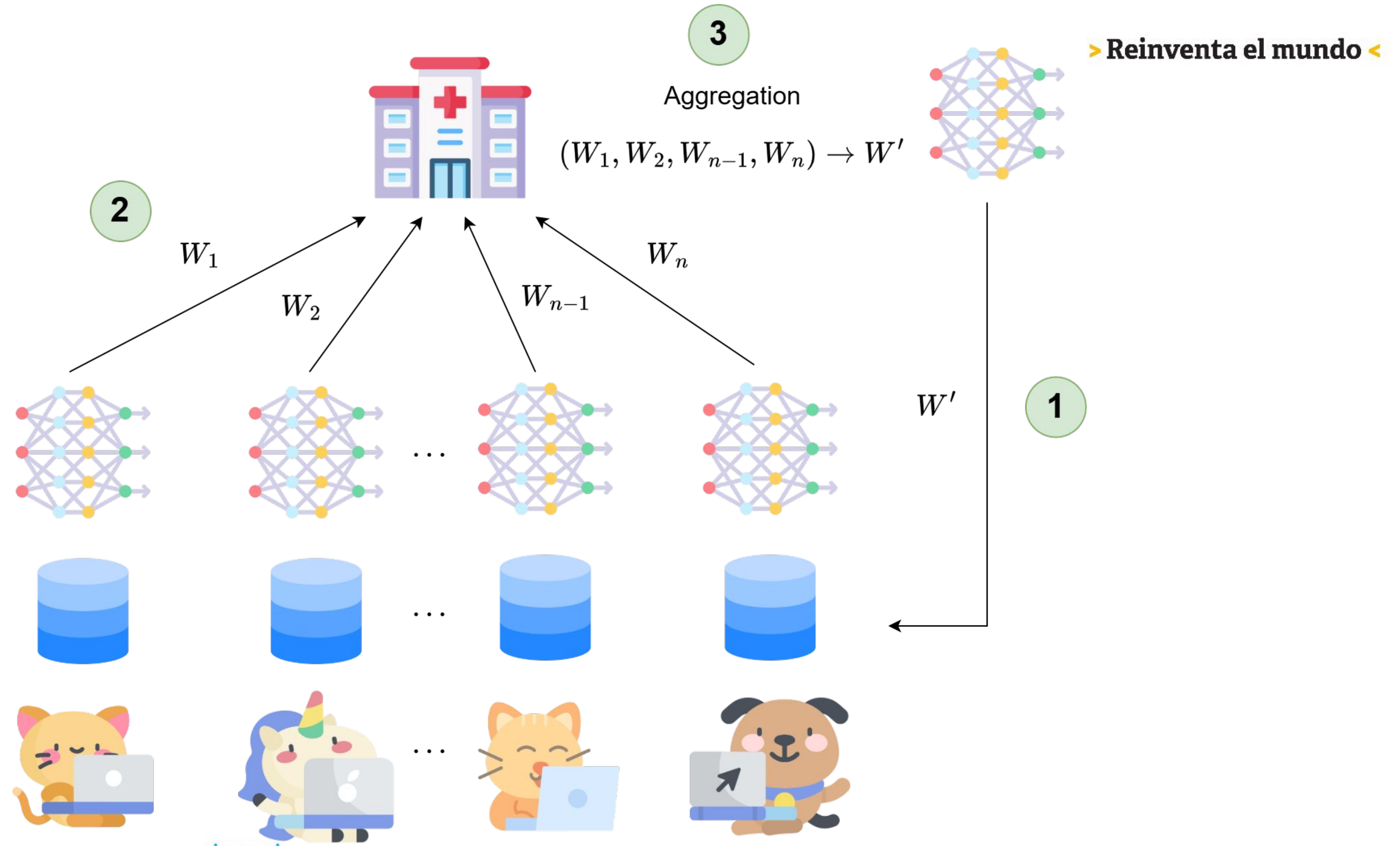


Figure 1: Traditional Federated Learning framework. (1) Model initialization; (2) Local model training and upload; (3) Global model aggregation and update.

Problem and Hypothesis

Problem

- Federated Learning (FL) struggles with **data heterogeneity** (non-IID data).
- Unknown how modern vision models behave under these conditions in medical imaging.
- Lack of benchmarks limits model selection for real deployments.

Hypothesis

- DeiT and Mamba will outperform CNNs under skewed FL.
- Their global context modeling helps deal with client heterogeneity.

Motivation

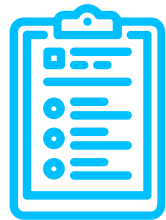
Computational Motivation

- FL performance drops under non-IID (skewed) conditions.
- Need to compare new vision architectures (DeiT, Mamba, ConvNeXt) vs CNNs in FL.
- Benchmarks help identify robustness to feature/label skew in medical images.

Practical Motivation

- Privacy laws (HIPAA/GDPR) prevent data sharing in healthcare.
- Medical data is naturally heterogeneous across institutions.
- Goal: Find models that stay reliable in real-world federated hospital settings.

2.



Frame *work*

2.1.



Federated *Learning*

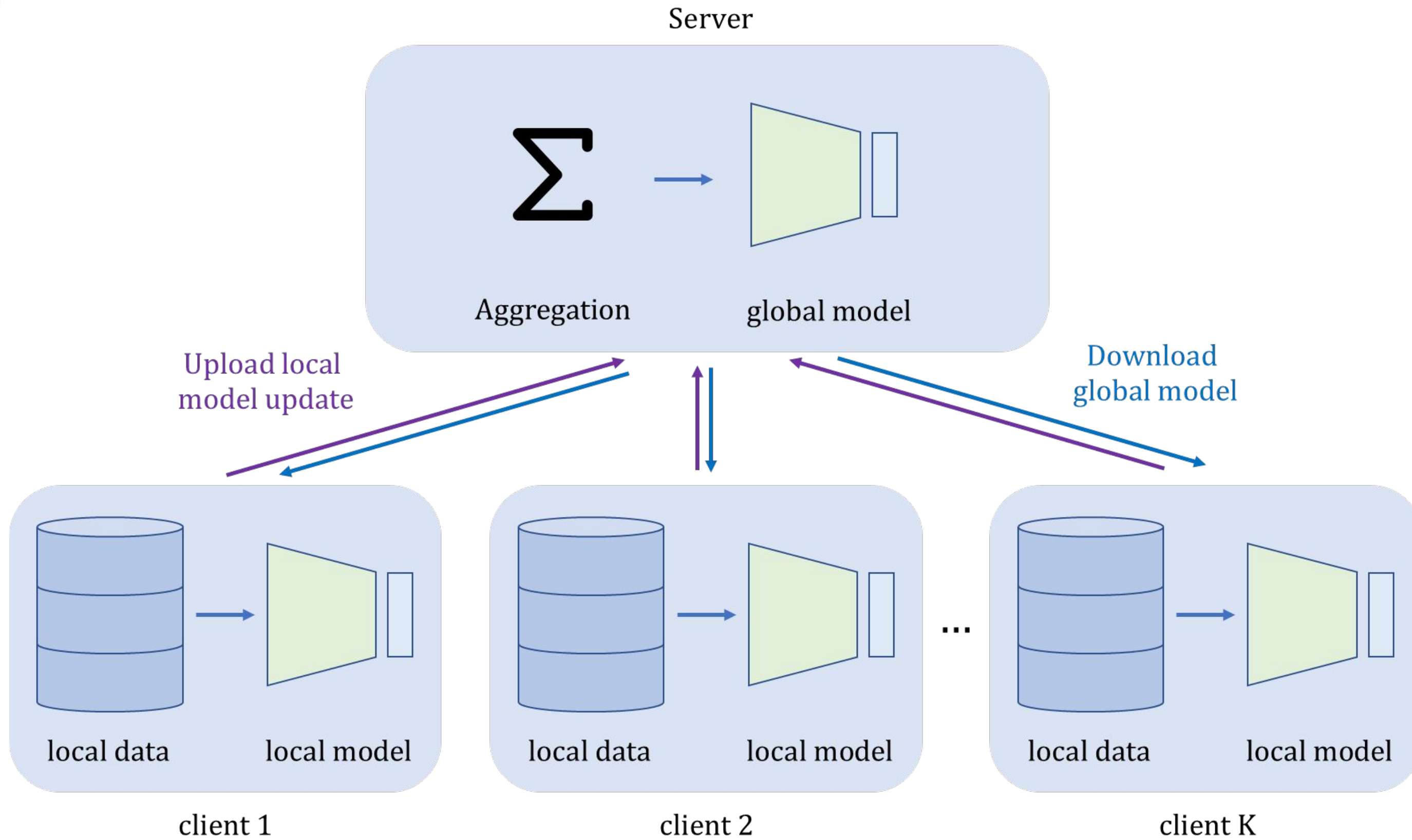


Figure 2: Traditional FL framework.

2.2.



Data *heterogeneity*

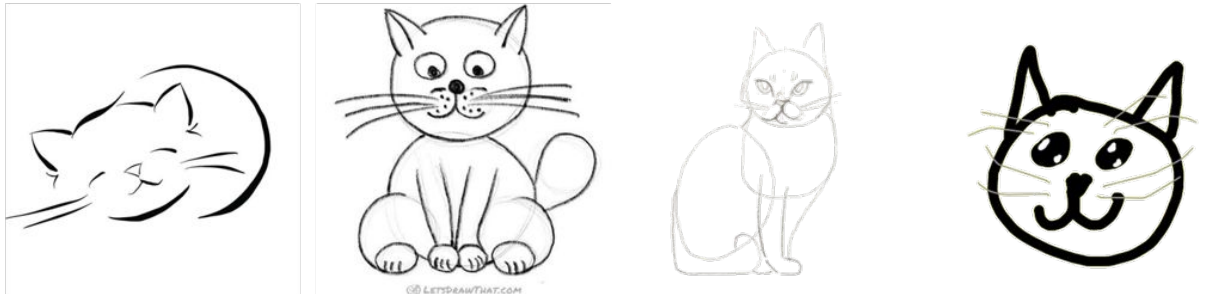
	Image	Label
Client i		cat
Client j		
Client k		

Figura 3: Illustration of feature distribution skew in data heterogeneity.

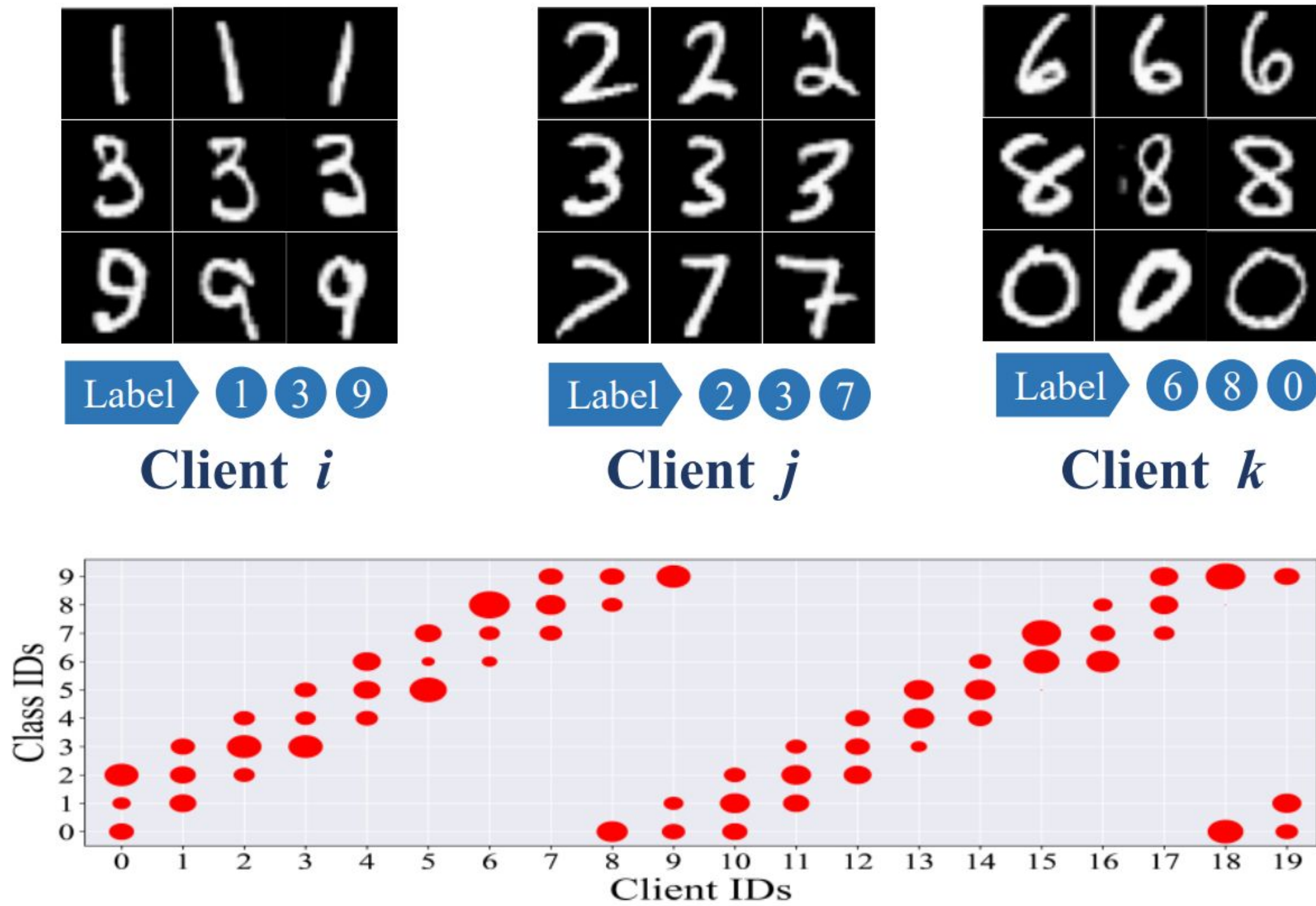


Figura 4: Illustration of label distribution skew in data heterogeneity.

2.3.



Vision *models*

Efficient *Net-B0*

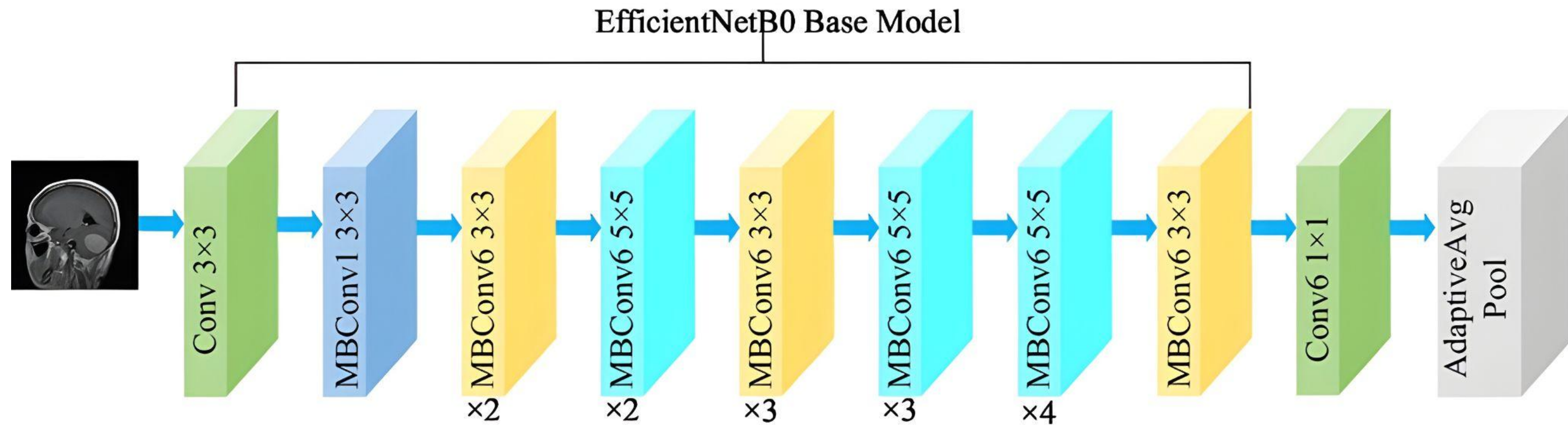


Figure 5: Architecture of the EfficientNetB0 Base Model used for Brain MRI Analysis.

Data-efficient Image Transformer (*DeiT*)

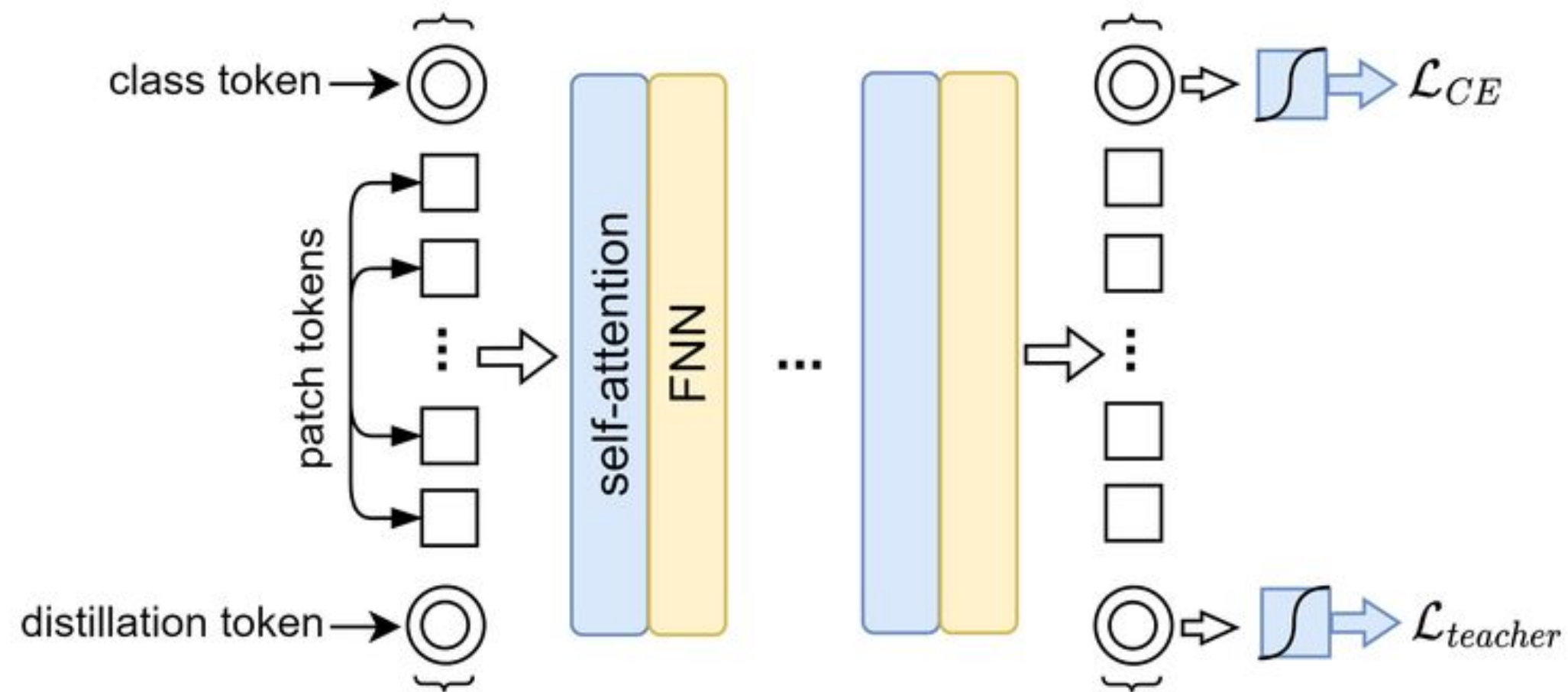


Figure 6: DeiT architecture illustrating the use of class and distillation tokens for self-attention and subsequent loss calculations.

ConvNeXt V1, V2

ConvNeXt V1 Block ConvNeXt V2 Block

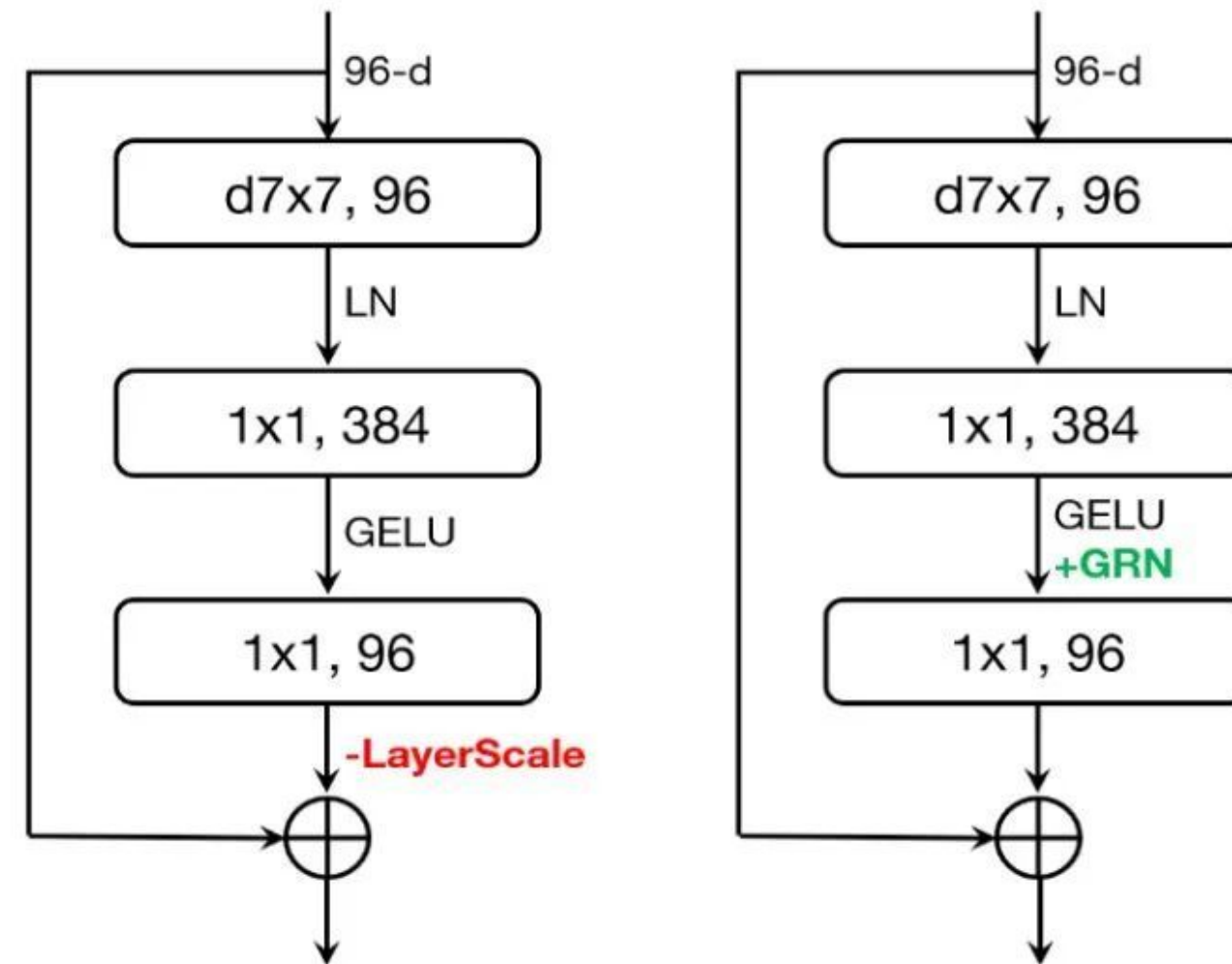


Figure 7: Comparison of ConvNeXt V1 and V2 blocks, highlighting the addition of Global Response Normalization (GRN) in V2 and the removal of LayerScale from V1.

Vision Mamba

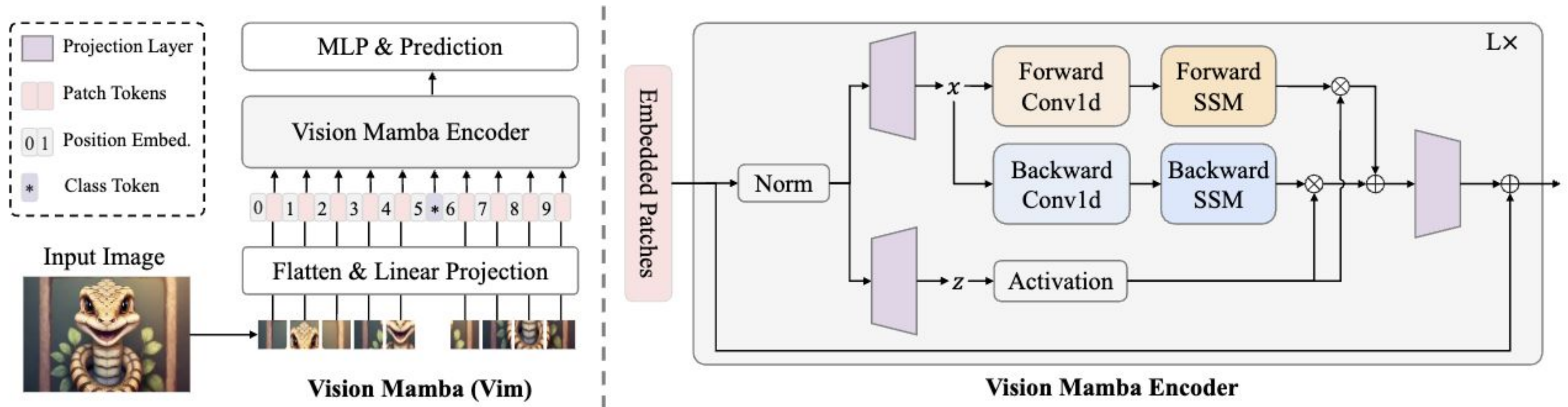


Figure 8: Overview of the Vision Mamba (ViM) architecture, detailing its patch embedding process and the internal structure of the Vision Mamba Encoder with forward and backward SSMs.

2.4.



Federated *methods*

Federated *Methods*

Traditional FL Methods

- **FedAvg:** Baseline method; highlights performance drop under data heterogeneity.
- **FedProx:** Adds regularization to handle client drift in non-IID data.
- **MOON:** Uses contrastive learning to align representations and reduce heterogeneity impact.

FL Domain Generalization Methods

- **FedSR:** Enhances generalization by improving local feature representations.
- **ADCOL:** Uses contrastive learning to boost global and local consistency.
- **FedIIR:** Combines representation learning and reweighting to adapt to non-IID clients.

3.



Related *works*

Related *works*

- **Rethinking Architecture Design for Tackling Data Heterogeneity in Federated Learning (2023):** It emphasizes how model architecture can impact FL performance under non-IID settings, directly aligning with my goal of benchmarking.
- **Label-Efficient Self-Supervised Federated Learning for Tackling Data Heterogeneity in Medical Imaging (2023):** This work addresses both label scarcity and data heterogeneity—two major challenges in medical imaging FL
- **Integrated approach of federated learning with transfer learning for classification and diagnosis of brain tumor (2024):** Combines FL with transfer learning to improve classification performance on medical images.
- **Distributed Federated Learning-Based Deep Learning Model for Privacy MRI Brain Tumor Detection (2024):** Demonstrates a privacy-preserving FL approach tailored for MRI data.

Related *works*

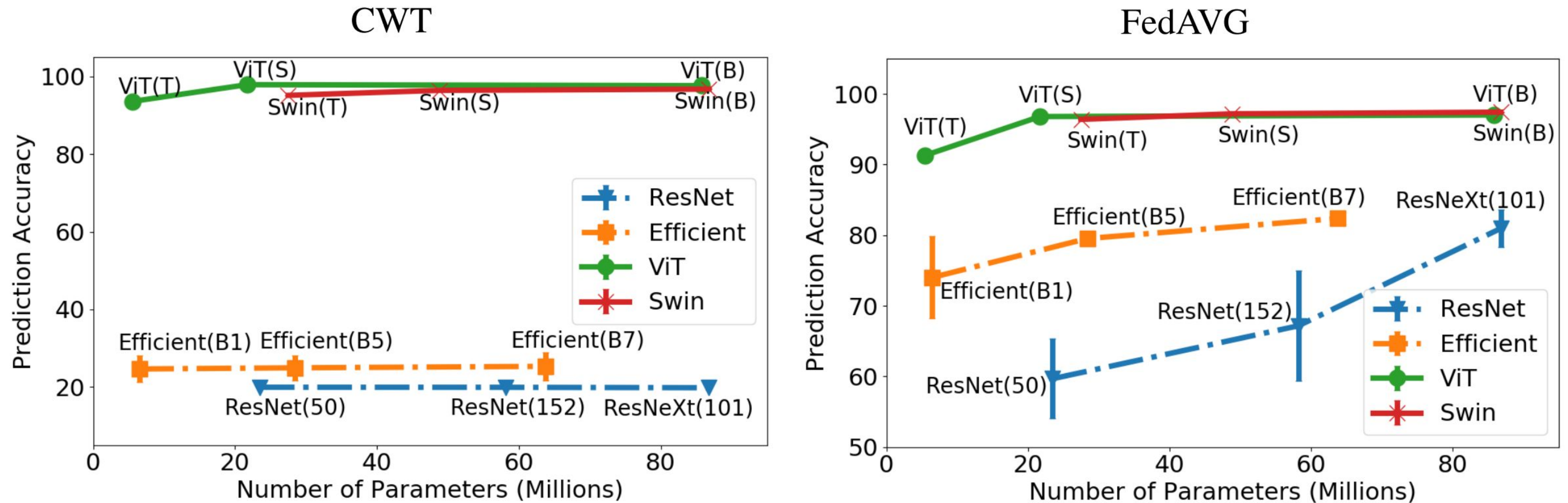


Figura 9: Rethinking Architecture Design for Tackling Data Heterogeneity in Federated Learning (2023). Prediction test accuracy on highly heterogeneous data partitions (Split-3) of CIFAR-10 dataset versus model size. Vision Transformers (ViTs and Swin Transformers) significantly outperform CNNs (ResNets and EfficientNets) on highly heterogeneous data partitions.

Related *works*

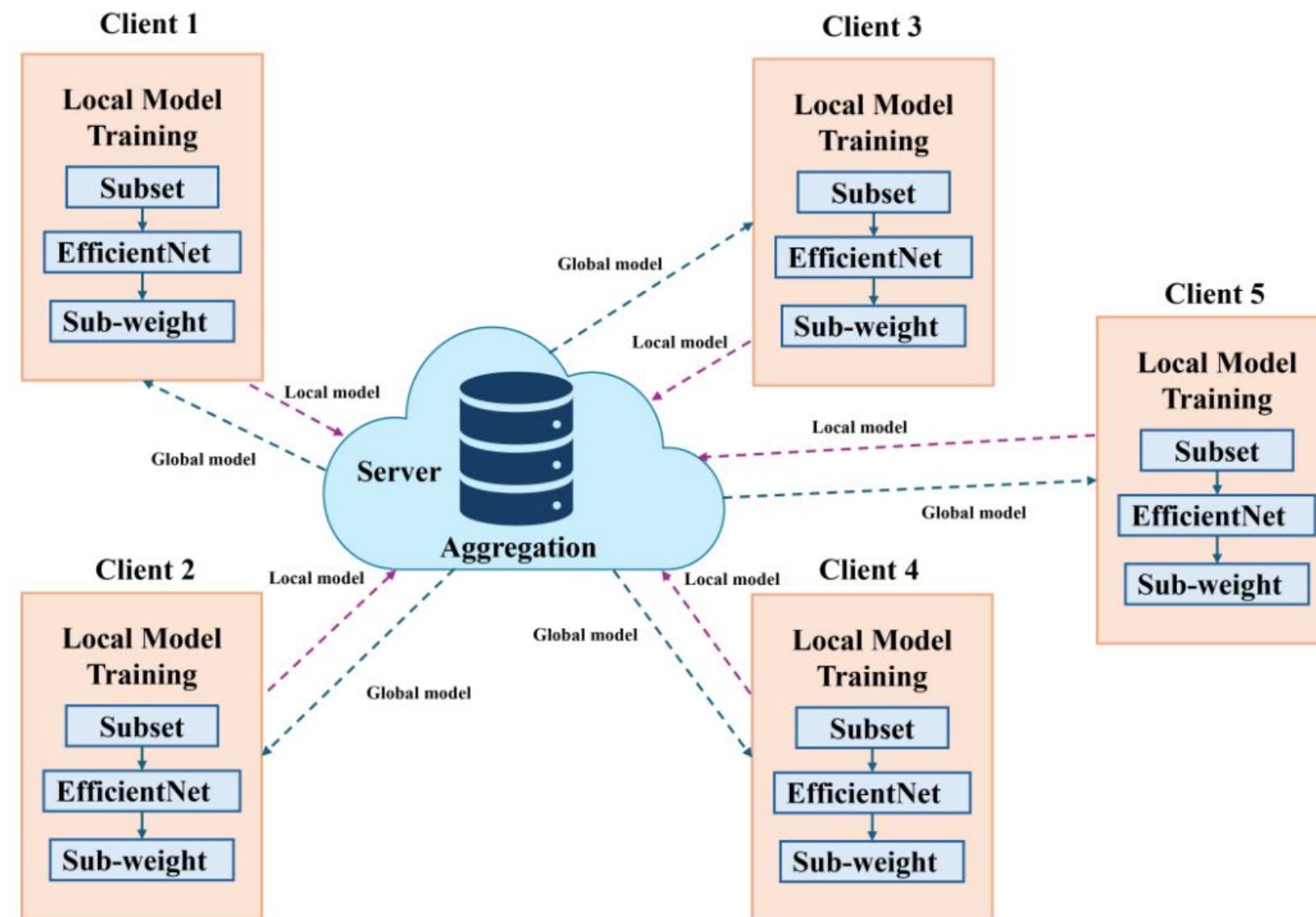


Figura 10: Distributed Federated Learning-Based Deep Learning Model for Privacy MRI Brain Tumor Detection (2024). The mechanism diagram of the proposed federated learning-based method.

4.



Method *ology*

Methodology

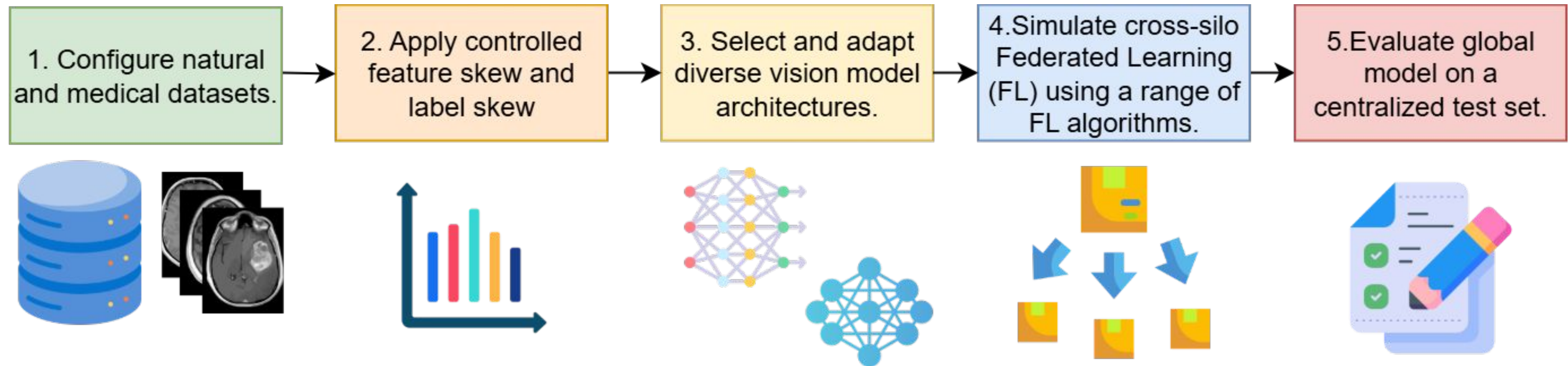


Figure 11: Overview of the thesis pipeline: datasets are prepared, models are selected, and cross-silo Federated Learning is simulated under feature and label skew. Models are trained for 100 rounds and evaluated on a centralized test set.

5.



Preliminary *results*

Implementation *details*

Cloud Infrastructure

Deployed experiments on an AWS EC2 g4dn.xlarge instance:

- GPU: NVIDIA Tesla T4 (16 GB VRAM)
- CUDA Version: 12.9
- Driver Version: 575.57.08
- Instance Specs: 4 vCPUs, 16 GiB RAM
- OS: Ubuntu 20.04 (base Conda environment)

Benchmark Framework

Utilized FL-Bench:

- A standardized benchmark suite for evaluating Federated Learning (FL) algorithms.
- Supports various non-IID data partitioning strategies, aggregation methods, and datasets.



Figure 12: PNY NVIDIA Tesla T4 Datacenter Card 16GB GDDR6 PCI Express 3.0 x16, Single Slot, Passive Cooling.

Vision *models*

Model name	Implementation	Pretrained weights	Number of parameters
Efficient Net B0	<code>torchvision.models.efficient_net_b0</code>	<code>EfficientNet_B0_Weights.DEFAULT</code>	5 288 548
DeiT-tiny	<code>timm.create_model('deit_tiny_patch16_224')</code>	<code>deit_tiny_patch16_224.fb_in1k</code>	5 000 000
ConvNeXt-tiny	<code>torchvision.models.convnext_tiny</code>	<code>ConvNeXt_Tiny_Weights.DEFAULT</code>	28 589 128
Vim-tiny	<code>vim_tiny_patch16_224_bimamba_v2_final_pool_mean_abs_pos_embed_with_midclstok_div2</code>	<code>Vim-tiny-midclstok</code>	7 000 000

Table 1: vision models selected for federated learning training. Each model is listed with its implementation source, pretrained weights (all on ImageNet-1k), and number of parameters. The selection includes convolutional architectures (EfficientNet, ConvNeXt), transformer-based models (DeiT), and a state-space model (Vim-tiny), which is based on Vision Mamba, a recent alternative to transformers for visual tasks.

Evaluation *metrics*

- Accuracy
- Precision
- Recall (Sensitivity)
- F1 Score

		POSITIVE	NEGATIVE
ACTUAL VALUES	POSITIVE	TP	FN
	NEGATIVE	FP	TN

$$Precision = \frac{TP}{TP + FP} \quad Recall = \frac{TP}{TP + FN}$$

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

CIFAR-10 Dataset

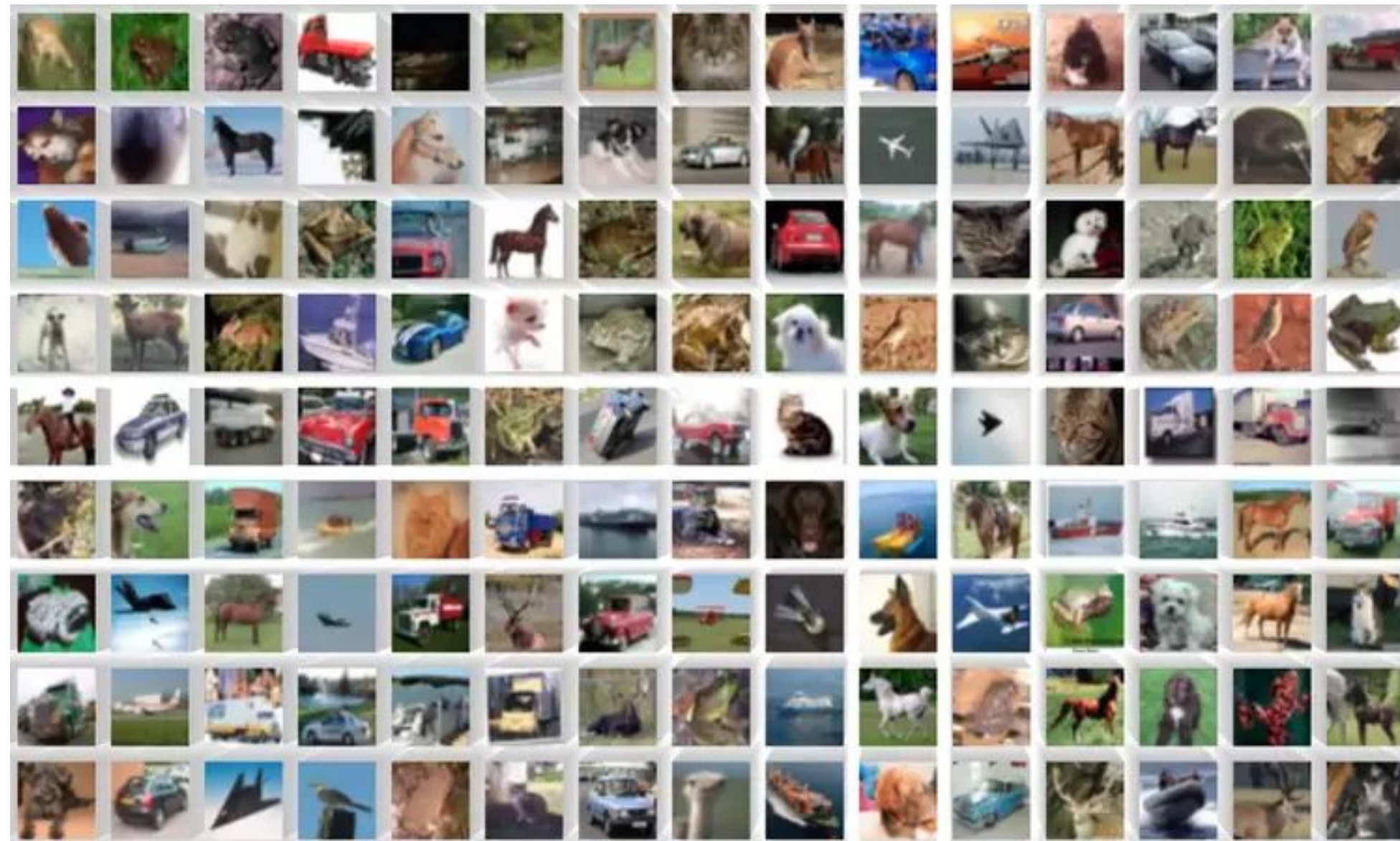


Figure 13: 60000 32x32 colour images in 10 classes, with 6000 images per class. There are 50000 training images and 10000 test images.

<https://www.cs.toronto.edu/~kriz/cifar.html>

CIFAR 10 *IID partition*

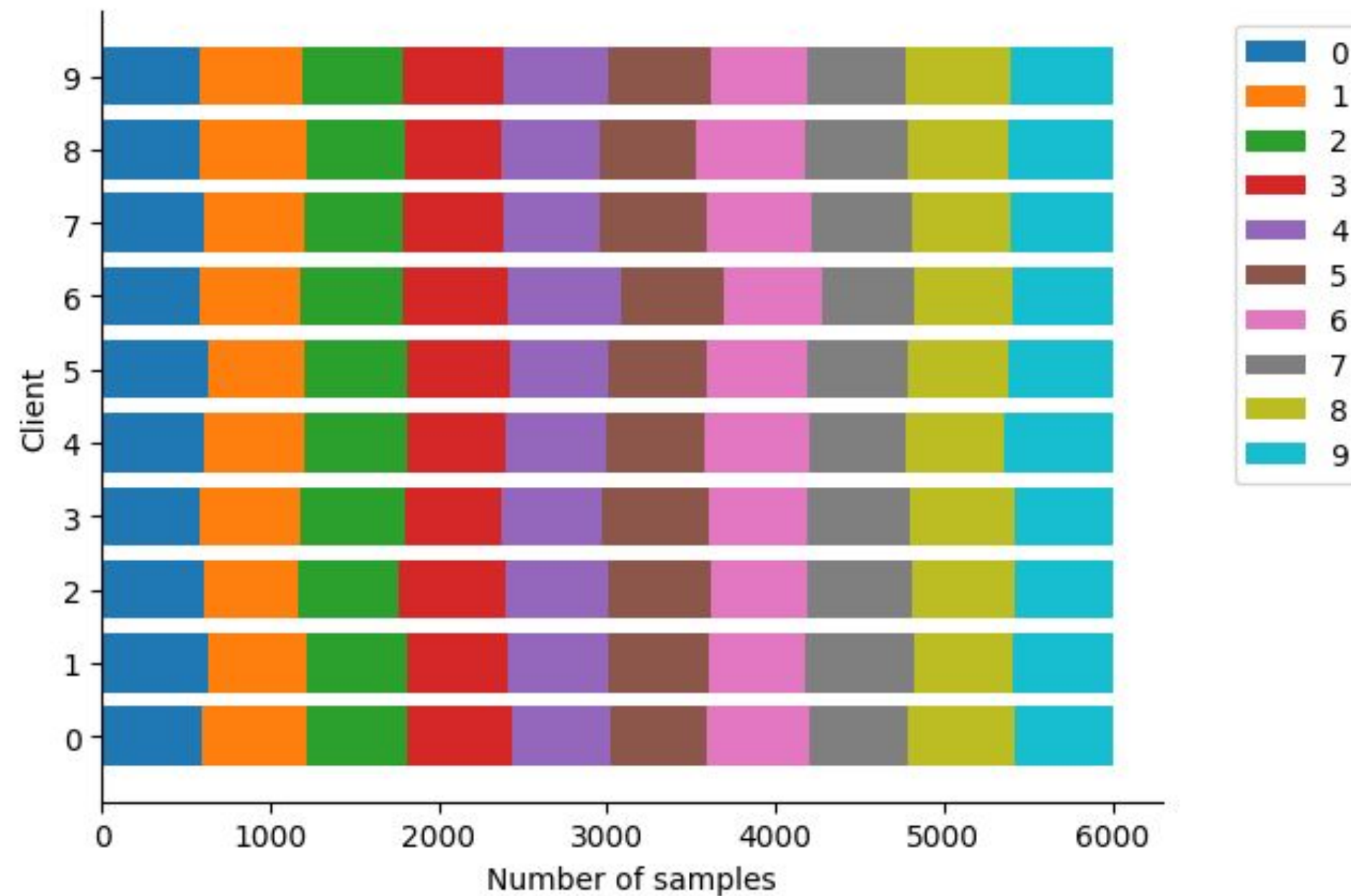
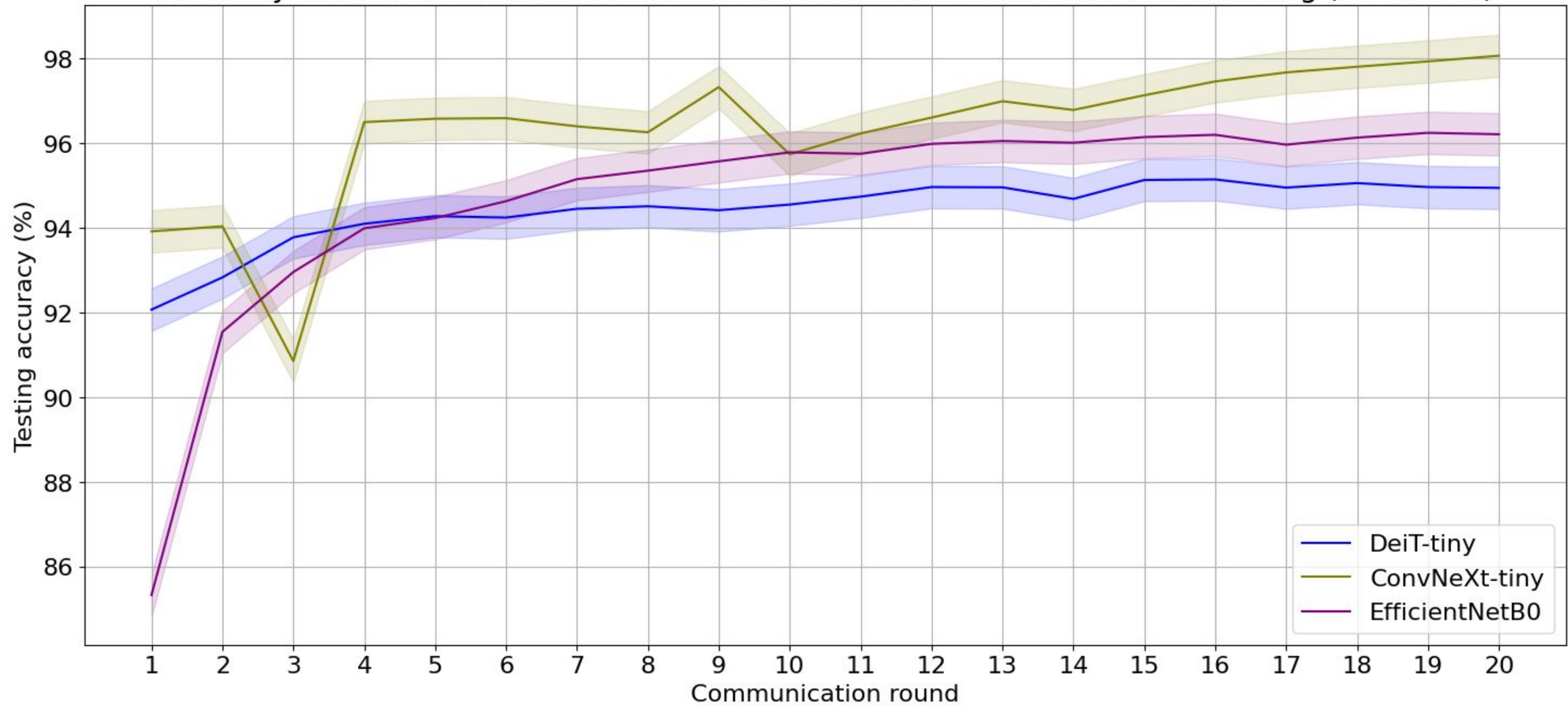


Table 14: Data distribution of the CIFAR-10 dataset among 10 clients. Each horizontal bar represents a client, and the different colored segments indicate the number of samples per class. The partition is balanced in size but non-IID in class composition.

Accuracy of Pretrained Vision Models on CIFAR-10 dataset in Federated Learning (10 Clients)



Key Observations:

1. ConvNeXt-Tiny achieved the highest accuracy (~98.5%) with fast and stable convergence, making it the most effective in this setting.
2. EfficientNetB0 showed competitive performance (~96.5%) but with more fluctuation, especially in early rounds.
3. DeiT-Tiny had the lowest performance (~95%) but remained stable, suggesting robustness, albeit with slower learning.

Brain Tumor MRI Dataset

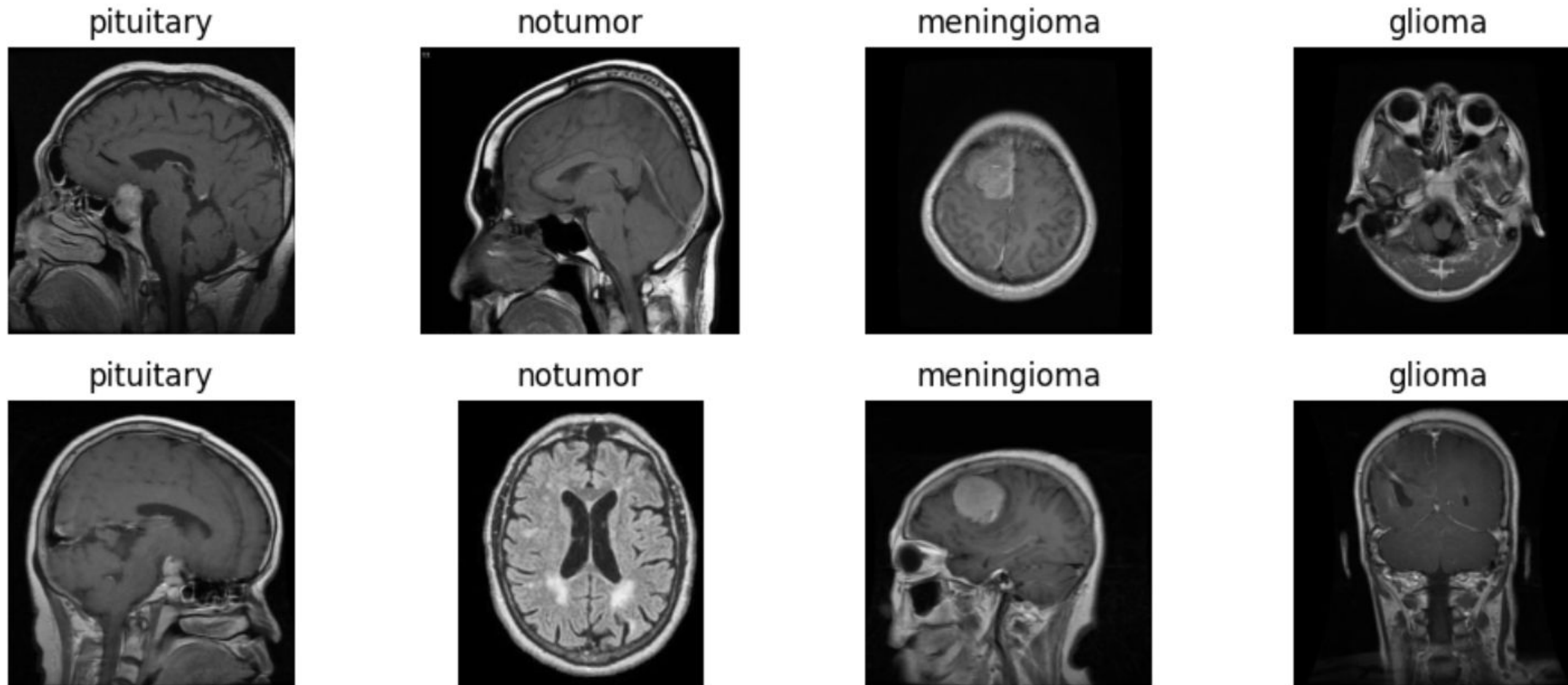


Figure 15: Brain Tumor MRI Dataset samples (Kaggle, Masoud Nickparvar). Collection of 7023 T1-weighted human brain MRI images (512×512px).

<https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>

Brain Tumor MRI Dataset *IID and Non-IID partitions*

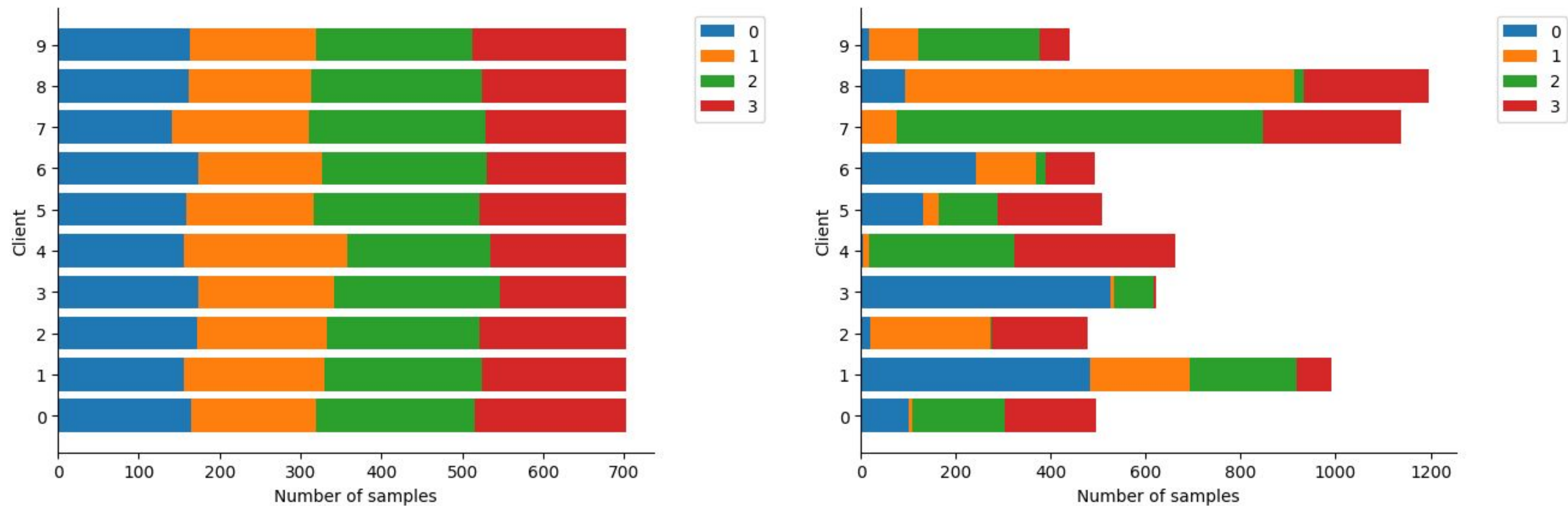
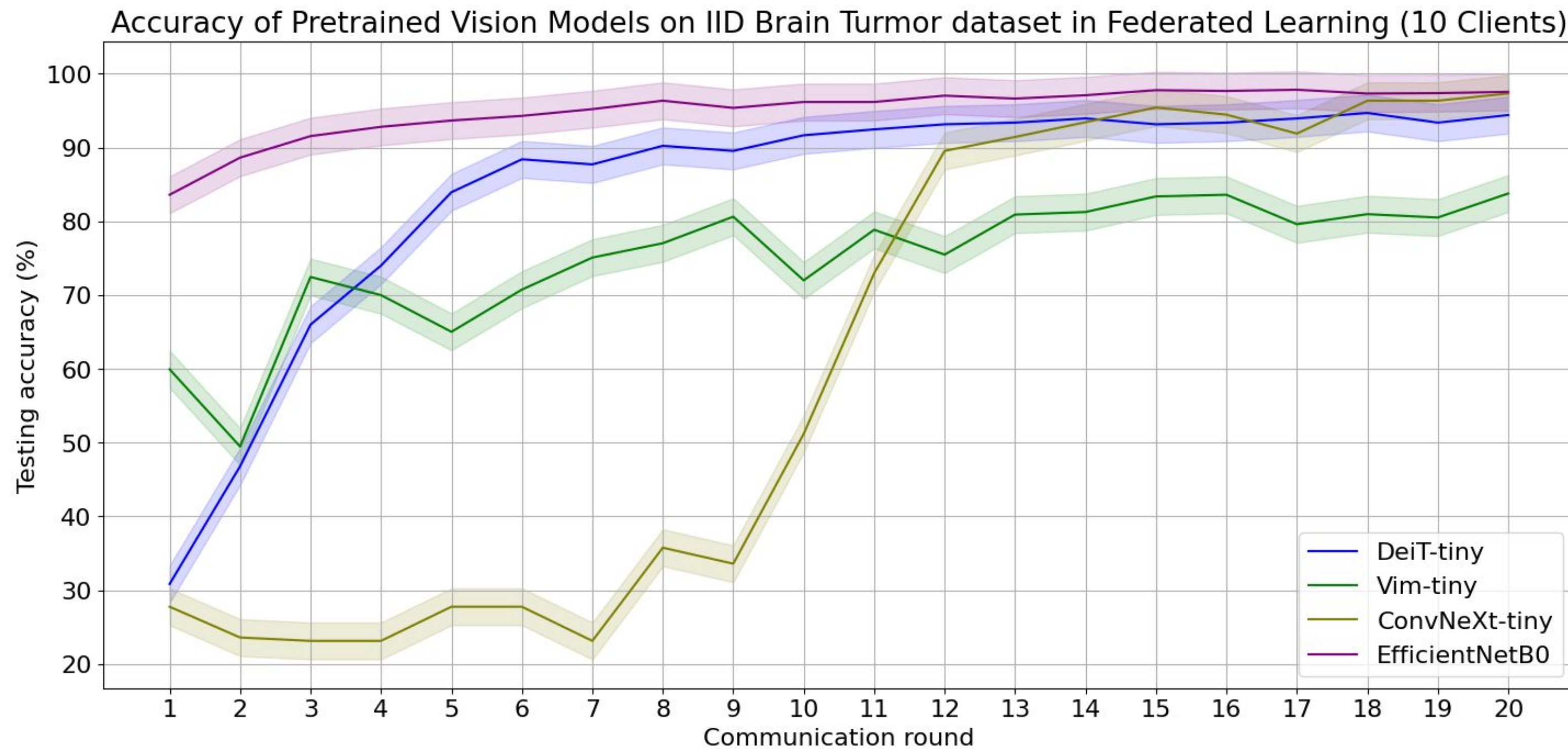
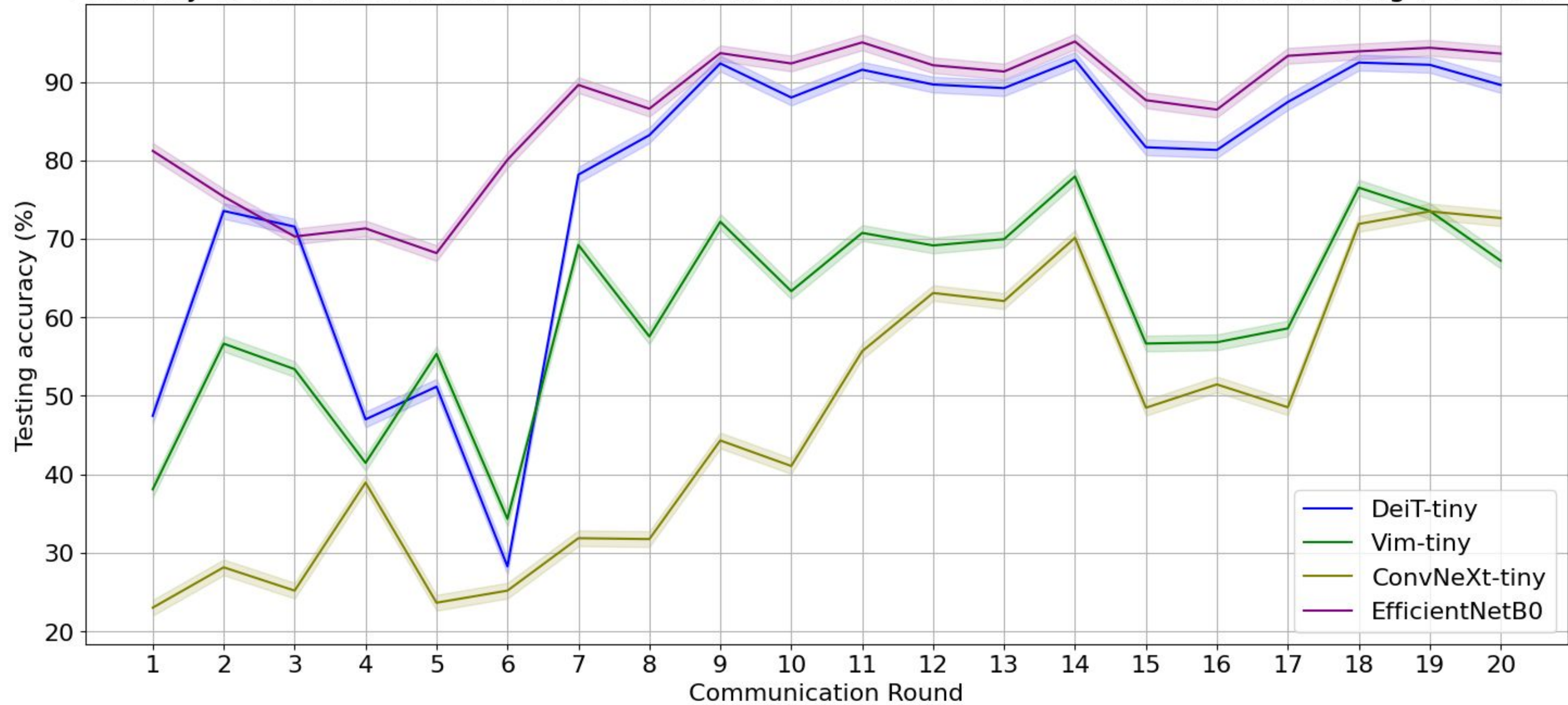


Figure 16: The image shows the IID (left) and Non-IID (right) data partitions of a Brain Tumor MRI Dataset across multiple clients, illustrating balanced and unbalanced class distributions, respectively.



Accuracy of Pretrained Vision Models on Non-IID Brain Tumor dataset in Federated Learning (10 Clients)



Key Observations

Model Performance Hierarchy

- EfficientNetB0 demonstrates superior and most stable performance across both datasets
- DeiT-tiny shows strong performance but with higher variance
- Vim-tiny and ConvNeXt-tiny exhibit lower overall accuracy

Dataset-Specific Patterns

- IID Dataset: All models show steady convergence with EfficientNetB0 reaching ~97% accuracy
- Non-IID Dataset: Higher volatility and instability, particularly affecting transformer-based models (DeiT-tiny, Vim-tiny)

Brain Tumor MRI Dataset *Non-IID partition*

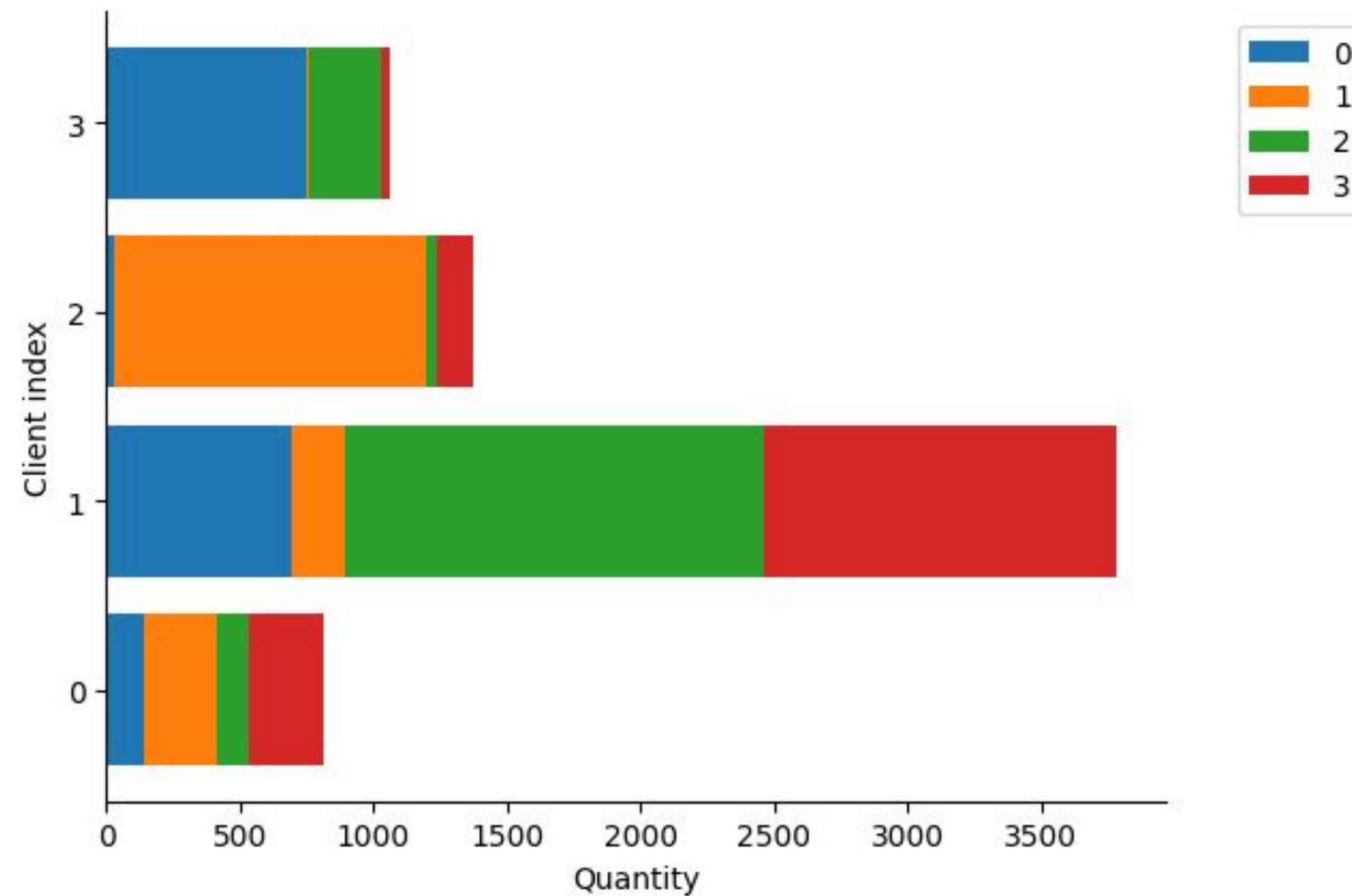
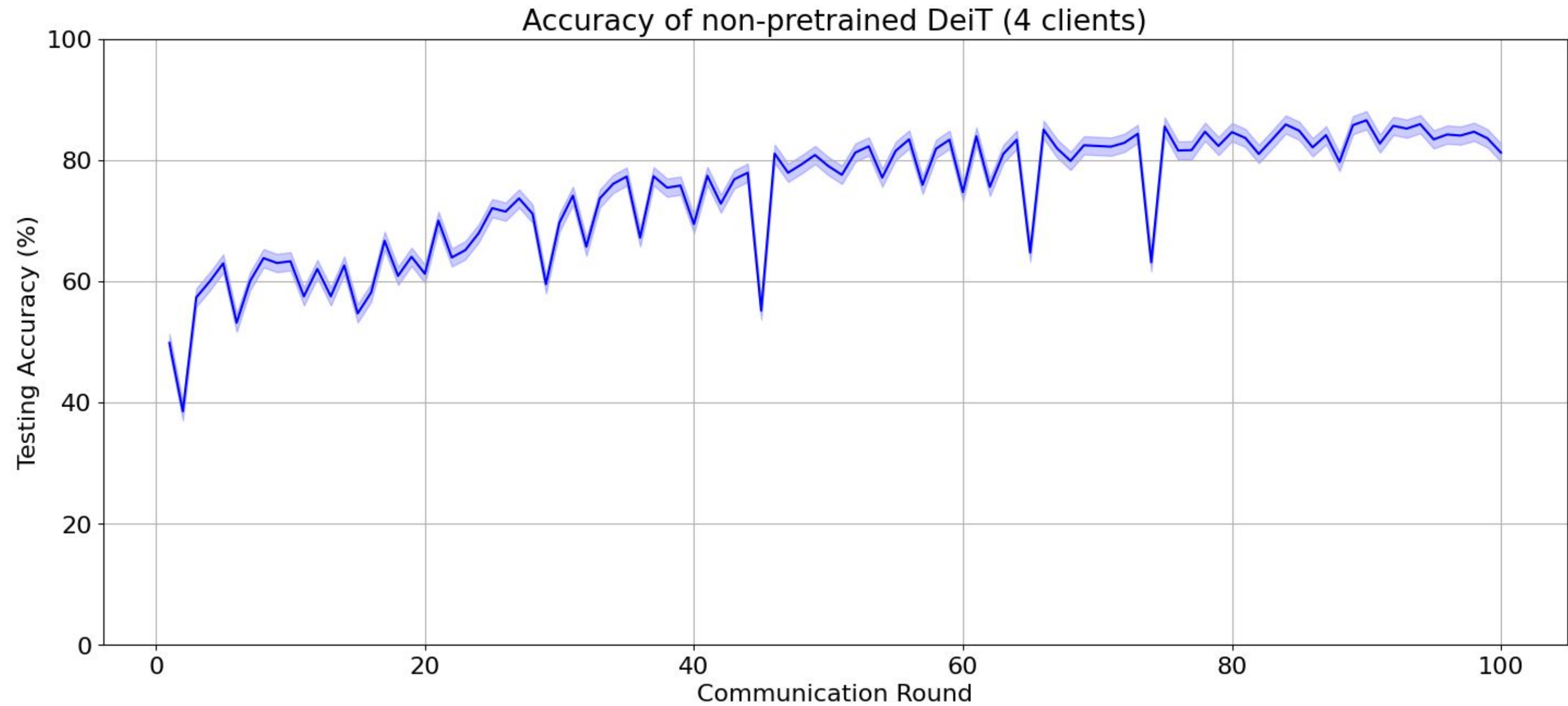
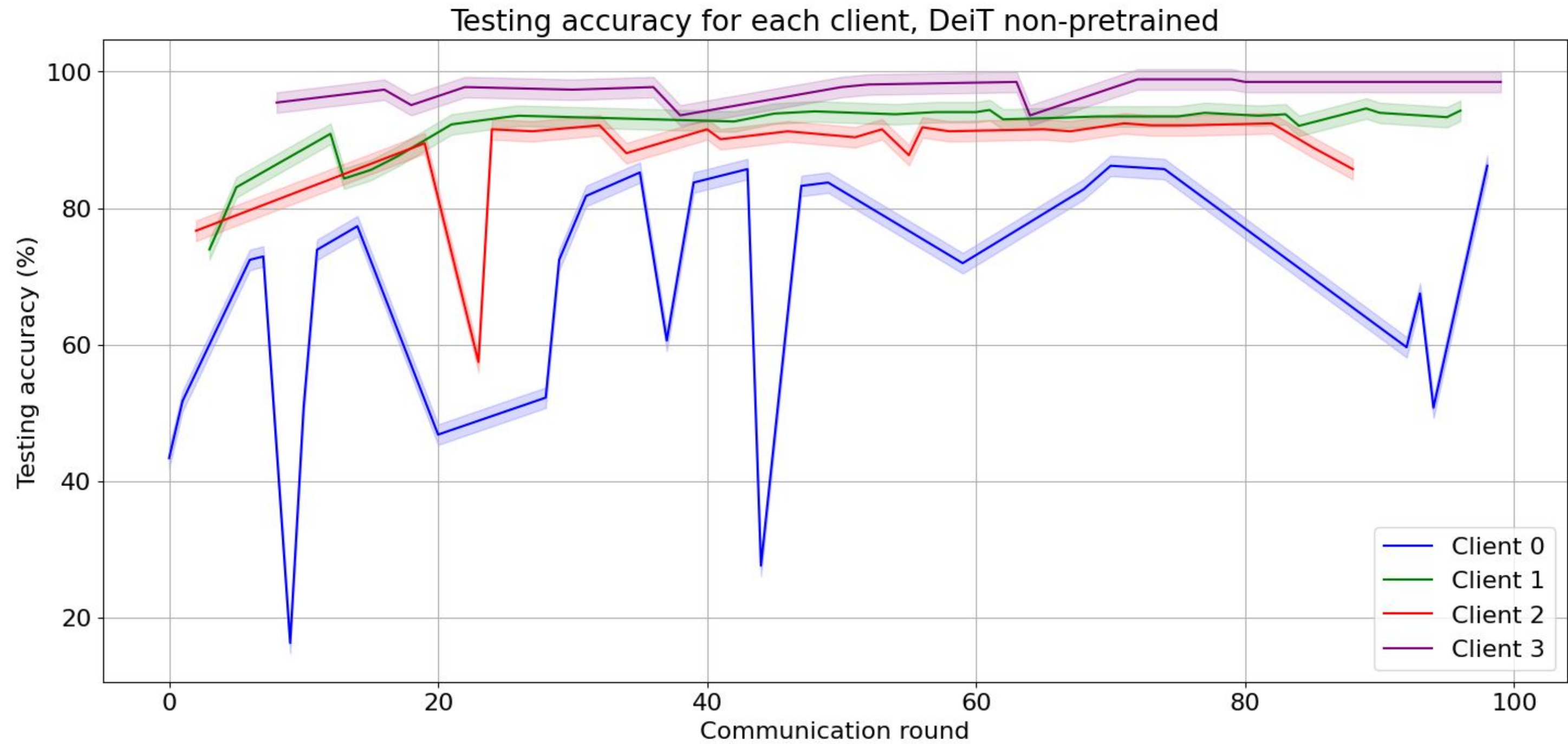


Figure 17: illustrates a non-IID (independent and identically distributed) partition of a Brain Tumor MRI Dataset, showing the quantity distribution of four categories (0, 1, 2, 3) across four different clients.





Key Observations

Client Performance Disparity

- Clients 1, 2, 3: Achieve consistently high accuracy (90-98%) with stable convergence
- Client 0: Shows severe instability and poor performance (20-85% range)

Convergence Patterns

- Stable Clients: Smooth upward trajectory reaching plateau around round 20-30
- Unstable Client: Extreme volatility with frequent dramatic drops (e.g., rounds 15, 40, 95)

DomainNet Dataset

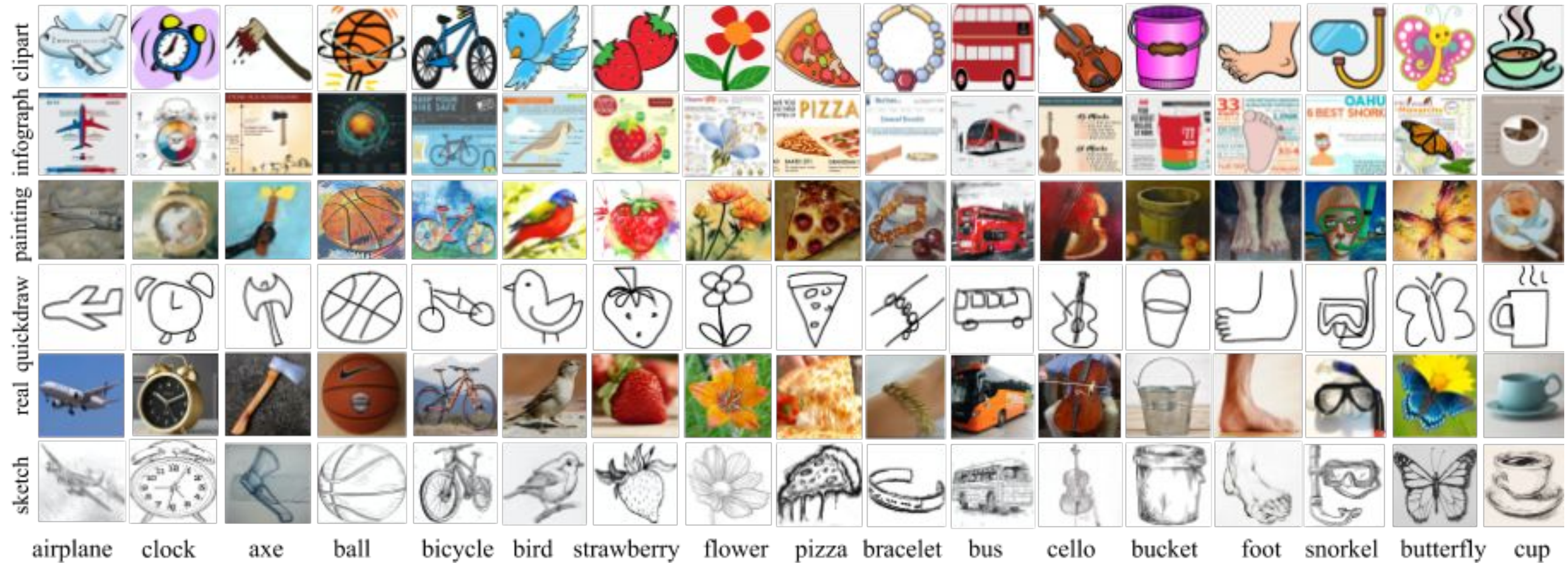


Figure 18: DomainNet Dataset samples. <https://ai.bu.edu/M3SDA/>

DomainNet *partition*

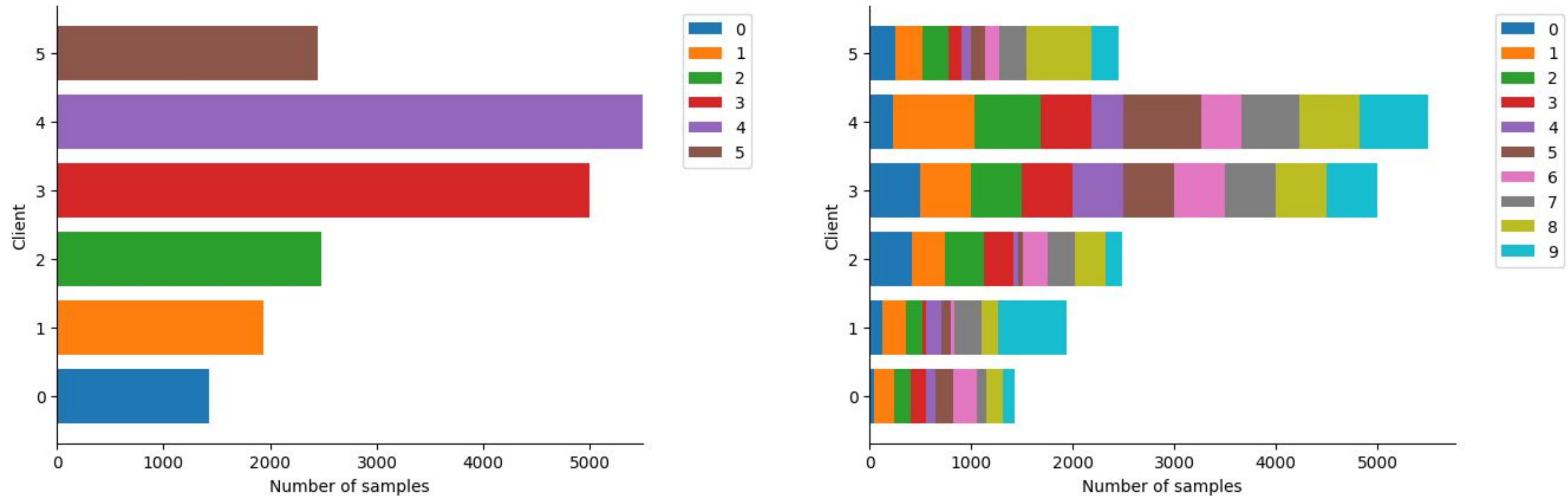
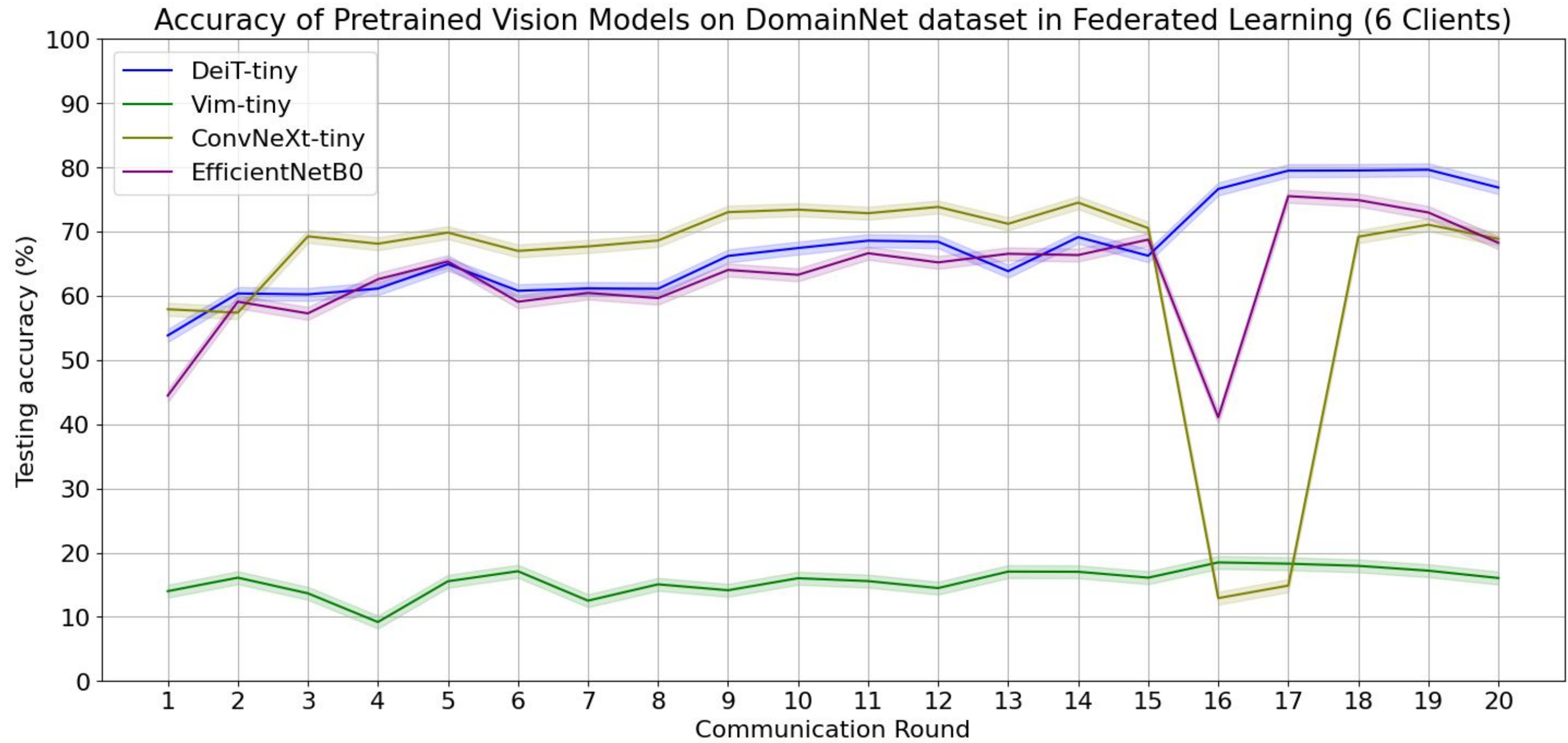


Figure 19: displays a "DomainNet partition" across six clients, with the left chart showing the total number of samples per client (each dominated by a single category), and the right chart detailing the distribution of ten different categories within each client's sample.



Key Observations

Model Performance Ranking

- DeiT-tiny: Best overall performance, reaching ~78% accuracy with steady improvement
- ConvNeXt-tiny: Second best, ~70% accuracy with moderate stability
- EfficientNetB0: Moderate performance (~65%) but shows catastrophic failure at round 16-17
- ViT-tiny: Consistently poorest performance, stagnating around 10-18%

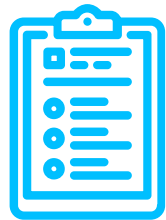
Training Stability

- DeiT-tiny: Most stable with consistent upward trend
- ConvNeXt-tiny: Relatively stable with minor fluctuations
- EfficientNetB0: Dramatic collapse at round 16-17, recovering by round 18
- ViT-tiny: Persistently unstable and low-performing

Future experiments

- Comprehensive FL Algorithm Evaluation
- Systematic Feature and label Skew Analysis
- Medical Dataset Evaluation
- Client Heterogeneity Analysis
- Scalability Experiments

6.



References

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